

Reward-Shaped Sequential Diagnostic Planning for Structured Wafer Defect Reasoning

Anonymous submission

Abstract

Wafer defect diagnosis in practical semiconductor manufacturing is inherently a partially observable sequential planning problem, where diagnostic decisions must be progressively refined under uncertain visual evidence and process-aware constraints. Rather than treating wafer inspection as independent classification, we formulate diagnosis as a compact sequential policy-learning problem with a fixed five-stage diagnostic trajectory. We present *WaferMapVQA*, a lightweight vision-language framework that casts wafer-map inspection as reward-shaped sequential diagnostic planning. The framework decomposes diagnosis into five discrete decision stages covering anomaly detection, defect classification, spatial localization, appearance description, and process-informed analysis. Each stage uses a four-option action space, producing a compact and auditable diagnostic policy rollout. For post-training, supervised fine-tuning (SFT) first stabilizes the policy over the structured action schema, and Group Relative Policy Optimization (GRPO) then improves the policy through three rule-based rewards: format validity, stage-weighted accuracy, and a domain-logic penalty that discourages inconsistent downstream defect reasoning on normal wafers. Experiments on a benchmark derived from the public WM-811K dataset show that the sequential SFT→GRPO setting achieves the best aggregate question-level accuracy of 92.43%. It also reduces false alarms from 11,995 under Pure GRPO to 3,293 while preserving a compact image-level five-action diagnostic trajectory. These results suggest that low-budget reward-shaped policy learning can turn a small open-source vision-language model into a practical diagnostic planning policy for semiconductor manufacturing.

Introduction

Wafer-map defect analysis is fundamental to semiconductor manufacturing because spatial defect patterns often reveal equipment conditions and yield instability. Our key insight is that reliable semiconductor diagnosis is fundamentally a sequential planning problem rather than an isolated classification task. Engineers first decide whether an abnormality exists, then refine a defect hypothesis through classification, localization, morphology interpretation, and process-aware diagnosis. Each diagnostic stage changes the downstream hypothesis space and therefore alters the subsequent decision policy. This dependency structure naturally resembles hierarchical diagnostic planning under uncertainty. Conventional approaches mainly rely on Convolutional Neural Net-

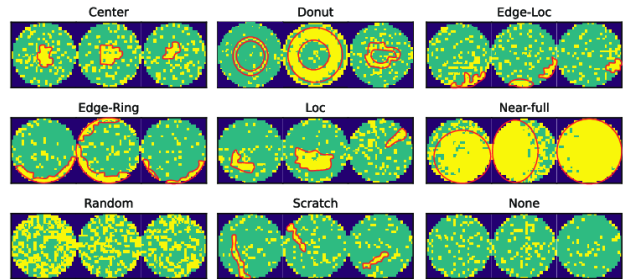


Figure 1: Representative wafer defect maps from the WM-811K dataset (Wu, Jang, and Chen 2014). The examples illustrate diverse spatial defect patterns, including center-dominant, ring-shaped, edge-localized, and scattered anomalies. Such regular morphology motivates structured and explainable diagnostic reasoning beyond conventional image-level classification.

works (CNNs) or clustering-based methods to classify patterns such as *Center*, *Donut*, and *Edge-Ring* (Chen et al. 2023; Saqlain et al. 2022; Wu, Jang, and Chen 2014). Although effective for pattern recognition, these methods remain largely black-box predictors and provide limited support for structured and auditable diagnostic interpretation. Representative examples from the public WM-811K wafer-map dataset (Wu, Jang, and Chen 2014), a large-scale benchmark for wafer defect pattern recognition, are shown in Fig. 1.

Recent multimodal large language models have enabled closer integration of visual perception and language reasoning. For example, *WamGLM* explores multi-turn querying for wafer-map analysis (Gu et al. 2025). More broadly, structured reasoning and reinforcement-based alignment have shown promise for improving reasoning stability through explicit intermediate steps (Shen et al. 2025; Pan et al. 2025). However, directly applying such methods to wafer inspection remains challenging, because reliable semiconductor diagnosis requires visual recognition, spatial grounding, manufacturing-aware interpretation, and deployment-oriented reliability.

To address this gap, we propose *WaferMapVQA*, a lightweight and deployable vision-language framework for

reward-shaped sequential diagnostic planning. Rather than treating wafer inspection as a closed-set classification task, WaferMapVQA reformulates diagnosis as a five-step diagnostic policy rollout—anomaly detection, classification, localization, description, and process-informed analysis—each implemented as a discrete four-option action to enable explicit and auditable supervision of intermediate decisions.

Our goal is not to pursue larger model scale, but to study whether a small open-source VLM can be transformed into a practical wafer-diagnosis assistant through low-budget post-training. Specifically, we construct a structured benchmark from WM-811K, use Supervised Fine-Tuning (SFT) to stabilize the action schema, and then apply Group Relative Policy Optimization (GRPO) to improve answer validity and deployment-oriented reliability. For the PRL setting, our main contributions are:

- We reformulate wafer inspection as a partially observable sequential diagnostic planning task, extending wafer-map analysis beyond single-label classification.
- We show that low-budget SFT→GRPO reward-shaped policy learning can substantially improve a small open-source VLM for structured semiconductor diagnosis.
- We evaluate not only aggregate question-level accuracy, but also deployment-oriented image-level indicators, including 5/5 correctness, 0/5 correctness, fatal misses, and false alarms.

Related Work

Wafer Map Defect Analysis

Wafer-map defect analysis is central to semiconductor yield management, as spatial defect patterns often indicate process excursions and yield instability. Early studies relied on rule-based methods and handcrafted descriptors, while later work adopted CNN-based models for wafer-pattern recognition (Chen et al. 2023; Saqlain et al. 2022; Wu, Jang, and Chen 2014). Although effective for classification, these approaches remain largely limited to single-label prediction and provide limited transparency in linking observed morphology to actionable process decisions.

Vision–Language Models for Industrial Analysis

Recent vision–language models (VLMs) have enabled closer integration of visual perception and language reasoning in industrial settings. In wafer analysis, *WamGLM* supports multi-turn dialogue and improves representation learning through prototypical contrastive learning (Gu et al. 2025). Related CLIP-guided anomaly-detection work has explored vision–language semantic alignment for few-shot defect detection (Lin and Lai 2025), while recent representation-learning approaches further improve anomaly modeling via quantized or structured latent spaces (Lin and Lai 2024). More broadly, reasoning-oriented post-training methods suggest that reinforcement-based alignment can improve stability and interpretability in vision–language systems (Shen et al. 2025; Pan et al. 2025). However, most existing studies focus on general multimodal reasoning or isolated anomaly detection rather than structured and deployment-oriented semiconductor diagnosis.

Sequential Planning and Reinforcement-Aligned Reasoning

Planning and reinforcement learning emphasize structured decision making, policy improvement, and reward-shaped behavior under uncertainty. Unlike conventional wafer classification pipelines, our formulation explicitly models diagnosis as a sequential decision trajectory under partially observable evidence. Although wafer inspection is often treated as static classification, practical fab diagnosis follows a sequential workflow: detect whether an abnormality exists, refine the defect hypothesis, localize the pattern, describe the morphology, and decide whether the observation suggests a process-relevant issue. This perspective is closely related to planning under uncertainty, policy shaping, and reinforcement-guided reasoning. It motivates our PRL-oriented formulation of wafer-map diagnosis as a compact diagnostic planning problem, where each stage corresponds to a discrete diagnostic action and GRPO provides lightweight reward-shaped policy learning for improving diagnostic consistency.

Research Gap

Existing wafer-inspection methods mainly optimize closed-set classification accuracy, whereas reasoning-oriented VLMs primarily target general multimodal reasoning. What remains underexplored is the intersection of these two directions: a lightweight, structured, and deployable framework that links perception-stage evidence to downstream diagnostic interpretation under an auditable protocol. This gap motivates the present study.

Scope and positioning Our goal differs from that of conventional wafer-map classification methods on WM-811K. Rather than replacing specialized single-label classifiers, we study a structured diagnostic setting that jointly evaluates anomaly detection, pattern recognition, localization, appearance characterization, and process-informed interpretation under a fixed five-question protocol. Accordingly, *WaferMapVQA* extends wafer-map analysis from isolated label prediction to auditable multi-stage diagnosis.

Sequential Diagnostic Planning Formulation

Fig. 2 summarizes the fixed five-stage diagnostic planning protocol and the two-stage SFT→GRPO reinforcement-alignment pipeline.

Partially Observable Diagnostic Policy Rollout

We study wafer-map images $I \in \mathbb{R}^{H \times W}$ with $H = W = 32$, where each cell encodes die-level defect statistics under low-resolution settings. No pixel-level masks or bounding boxes are provided. Beyond predicting a defect label, our goal is to produce an auditable diagnostic output that links spatial morphology to process-informed interpretation within a fixed and auditable action schema.

Motivated by practical yield-engineering workflows (*inspect* → *localize* → *describe* → *explain*), we formulate wafer diagnosis as a sequential diagnostic planning task under a fixed five-step decision protocol $Q = \{Q_1, \dots, Q_5\}$:

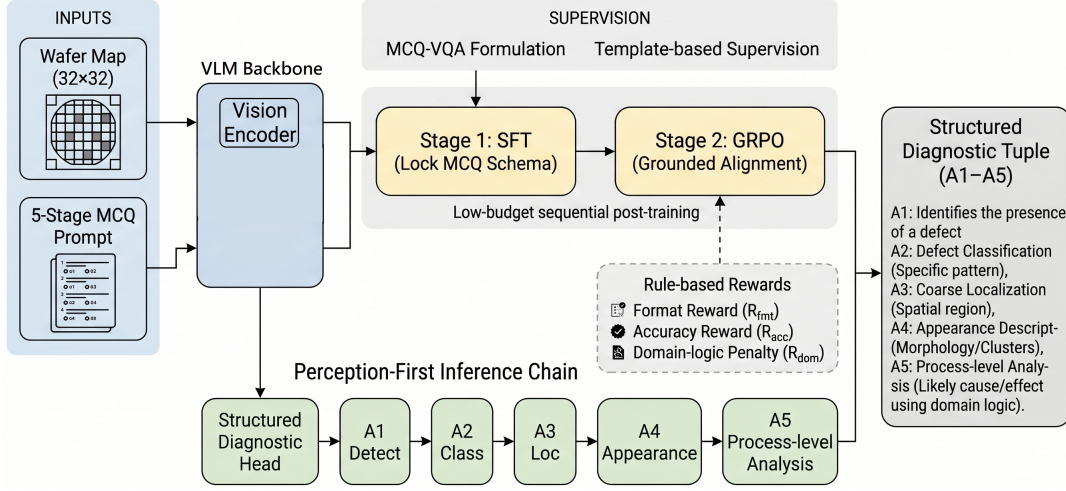


Figure 2: Sequential diagnostic policy rollout under a fixed five-stage planning protocol. The framework separates Q_1 as an anomaly gate and treats Q_2 – Q_5 as downstream defect-reasoning actions. Each stage updates the downstream diagnostic hypothesis space and influences subsequent policy decisions. Each decision instance takes a 32×32 wafer map and one stage-specific prompt as input; the five stage-wise actions are grouped into a structured diagnostic trajectory $A = \{A_1, \dots, A_5\}$. SFT stabilizes the fixed action schema, while GRPO improves policy validity and domain consistency using three rule-based rewards: format reward R_{fmt} , stage-weighted accuracy reward R_{acc} , and domain-logic penalty R_{dom} .

- Q_1 : anomaly detection,
- Q_2 : defect classification,
- Q_3 : coarse localization,
- Q_4 : appearance description,
- Q_5 : process-informed analysis.

Each step is instantiated as a four-option diagnostic action, and the policy must output exactly one option letter from $\{A, B, C, D\}$. Although the semantic meaning of the four options is template-specific and may vary across samples, the externally evaluated answer space remains fixed at four choices per question. The five-stage trajectory can be interpreted as a compact diagnostic policy rollout. The final prediction is a five-step diagnostic trajectory $A = \{A_1, \dots, A_5\}$, where $A_k \in \{A, B, C, D\}$ denotes the selected diagnostic action for Q_k . In both training and testing, Q_1 – Q_5 are asked as separate stage-specific decision instances; the five stage-wise decisions are then grouped into a diagnostic trajectory for image-level consistency analysis. This design enforces a perception-first planning protocol with explicit stage-wise supervision.

Template-based supervision For each defect category, we define standardized question–answer templates that map morphology cues to process-informed interpretations under the same four-option output constraint. These templates are used to construct consistent SFT pairs while keeping the evaluated answer format identical across SFT, GRPO, and final inference.

Scope of the supervisory targets Because the public WM-811K dataset does not provide verified fab-level root-cause annotations, the supervisory targets for later

stages are template-derived and should be interpreted as process-informed diagnostic surrogates rather than ground-truth causal labels. Accordingly, *WaferMapVQA* benchmarks structured diagnostic consistency and morphology-aware explanation under a fixed protocol, rather than direct causal discovery from lot-level manufacturing metadata.

Two-Stage Post-Training: SFT $\rightarrow$ GRPO

Our training follows a two-stage policy-learning recipe: SFT first stabilizes the structured diagnostic action space, and GRPO then improves policy validity and domain consistency. The task definition remains unchanged across both stages.

Stage 1: Supervised Fine-Tuning (SFT) SFT is used to (a) stabilize the five-stage action format, (b) teach the model to follow the fixed diagnostic planning protocol, and (c) provide an initialization policy for subsequent GRPO alignment. Each SFT instance contains one wafer map and one stage-specific MCQ, and requires the model to output a single option letter in a strict schema.

Stage 2: GRPO Reinforcement Alignment for Diagnostic Policy Learning We build the GRPO stage on a VLM-R1-style framework (Shen et al. 2025), following the broader DeepSeek-R1-style post-training paradigm for reasoning-oriented reinforcement alignment, where training samples and reward functions are prepared in JSON Lines (JSONL) format. For each input, the trainer samples a group of candidate completions and updates the policy using group-relative advantages. During sampling, the model may generate intermediate text before the final option letter; however, evaluation parses only the final option for each MCQ instance,

and the five stage-wise predictions are grouped into the externally reported five-answer tuple.

Reward design The total GRPO reward is defined in Eq. (1) as

$$R = R_{\text{fmt}} + R_{\text{acc}} + R_{\text{dom}}, \quad (1)$$

where R_{fmt} is a format reward, R_{acc} is a stage-weighted accuracy reward, and R_{dom} is a domain-logic penalty.

The format reward in Eq. (2) is

$$R_{\text{fmt}} = 0.2 \cdot \mathbb{I}[\hat{y} \in \{A, B, C, D\}], \quad (2)$$

where $\mathbb{I}[\cdot]$ denotes the indicator function and $\hat{y}$ is the parsed predicted option letter. Equation (2) rewards valid four-option MCQ outputs and discourages malformed responses.

The accuracy reward in Eq. (3) is

$$R_{\text{acc}} = w(Q_k) \cdot \mathbb{I}[\hat{y} = y^*], \quad (3)$$

where y^* is the ground-truth option letter and $w(Q_k)$ is the question-type-dependent weight for the current question in the five-stage protocol.

Finally, the domain-logic penalty in Eq. (4) is

$$R_{\text{dom}} = -0.5 \cdot \mathbb{I}[c = \text{none}] \cdot \mathbb{I}[j > 1] \cdot \mathbb{I}[\hat{y} \neq y^*], \quad (4)$$

where c denotes the wafer class label and j denotes the question index. Equation (4) penalizes defect-related follow-up reasoning errors on normal wafers.

Implemented reward functions In the released implementation, GRPO uses exactly three rule-based rewards: R_{fmt} , R_{acc} , and R_{dom} . No additional consistency or root-cause reward terms are used.

Key principle Across SFT and GRPO, we keep the same decision stages, option space, and output schema. GRPO improves action formatting, stage-sensitive diagnostic accuracy, and domain consistency without changing the task definition. From a planning perspective, the reward design acts as lightweight policy shaping that biases the diagnostic rollout toward semantically consistent trajectories. The reward terms jointly encourage valid action formatting, stage-aware decision accuracy, and cross-stage diagnostic consistency.

Implementation Details

Backbone and experimental settings Our main experiments are built on the instruction-tuned open-source 3B-scale VLM Qwen2.5-VL-3B-Instruct (Qwen Team 2025), where B denotes billions of parameters. To study how low-budget post-training changes diagnostic behavior, we compare four settings: (i) zero-shot inference, (ii) SFT on the 5% post-training pool, (iii) GRPO on a smaller nested subset drawn from that pool (denoted as Pure GRPO in our experiment naming), and (iv) a sequential SFT→GRPO pipeline. In addition, we include InternVL2.5-1B as a lightweight off-the-shelf reference model.

Codebase and implementation patches The proposed pipeline is implemented on top of the HuggingFace Qwen vision-language stack with several task-specific engineering patches. In addition to the standard generation backbone, we

use custom monkey-patched forward functions for mixed-modality execution and patched vision-tower components for stable windowed visual processing. These modifications ensure compatibility between the wafer-image input, MCQ protocol, and multimodal training pipeline.

Training stability and low-budget configuration Both SFT and GRPO are implemented under a low-budget training regime with optional LoRA adaptation, optional 4-bit or 8-bit quantization, configurable freezing of the vision tower, merger, and language model, and selective top- k layer unfreezing. In the GRPO stage, non-reentrant gradient checkpointing is used for stability under variable completion lengths in multimodal policy optimization, whereas SFT follows the standard supervised configuration. These implementation choices are designed to keep the training pipeline practical for small open-source backbones.

Sequential nested post-training The final reported setting is sequential rather than parallel: SFT is first trained on the full 5% post-training pool, and GRPO is then applied to a smaller nested training subset drawn from within that same pool. In code, this is implemented as reward-based continuation training on top of the supervised initialization rather than as an independent standalone branch.

Prompt protocol and action schema All wafer maps are represented as 32×32 grid images, and all prompts follow a fixed five-stage diagnostic protocol covering anomaly detection, defect classification, localization, description, and process-informed analysis. In the implemented setting, each stage is asked as a separate four-option MCQ instance with the same wafer map as input, and the model must output exactly one option letter from $\{A, B, C, D\}$ for that instance. The semantic content of the four options is instantiated per sample. For image-level evaluation, the five stage-wise predictions are grouped into a fixed five-answer tuple $\mathbf{A} = \{A_1, A_2, A_3, A_4, A_5\}$, where A_k denotes the selected option for question Q_k .

Dataset construction and split statistics The benchmark is organized into a 95% held-out test split and a 5% post-training pool derived from the public WM-811K wafer-map dataset. Tables 1 and 2 summarize the split statistics and image-level defect-class distribution, respectively. We use the full 5% pool (8,648 wafers) for SFT and a smaller nested subset of 5,880 wafers for GRPO. Final evaluation is conducted only on the 164,302-image held-out test split. Accordingly, the SFT and GRPO training sets are nested rather than additive, and the corresponding columns in Tables 1 and 2 should be interpreted separately from the test column.

Under the five-question protocol, the 5% SFT pool yields 43,240 supervised QA instances. From the nested GRPO wafer subset, the current GRPO run samples 8,648 training question instances, distributed as 1,730 samples for Q_1 – Q_3 and 1,729 samples for Q_4 – Q_5 . Final evaluation on the held-out test split contains 821,510 question instances.

Evaluation protocol Final evaluation is reported at two levels. First, we compute aggregate question-level accuracy across all evaluated question instances, together with per-

Table 1: Dataset split statistics. The GRPO subset is nested within the 5% SFT pool; image counts are reported at the wafer level and QA counts under the five-question protocol.

Split	Images	QA Instances
SFT pool (5%)	8,648	43,240
GRPO subset	5,880	8,648
Test (95%)	164,302	821,510

Table 2: Image-level defect-class distribution across the 5% SFT pool, nested GRPO subset, and 95% held-out test split.

Defect Class	SFT pool	GRPO subset	Test	Full
center	215	142	4,079	4,294
donut	28	20	527	555
edge_loc	259	175	4,930	5,189
edge_ring	484	330	9,196	9,680
loc	180	114	3,413	3,593
near_full	7	5	142	149
random	43	30	823	866
scratch	60	43	1,133	1,193
none	7,372	5,021	140,059	147,431
Total	8,648	5,880	164,302	172,950

stage accuracies for anomaly detection, defect classification, localization, description, and process-informed analysis. Second, we report stricter image-level consistency indicators, including 5/5-correct and 0/5-correct rates, where 5/5 denotes that all five evaluated answers for a wafer are correct simultaneously under the same evaluation protocol. We additionally report deployment-oriented error statistics, including fatal misses (FM) and false alarms (FA), to characterize model behavior under practical fab-oriented operating constraints.

Tables 1 and 2 summarize the split statistics and image-level defect-class distribution.

Experimental Results

Main Results and Deployment-Oriented Analysis

Table 3 reports the main stage-wise results on *WaferMapVQA*, and Table 4 summarizes the deployment-oriented error profile. Because wafer diagnosis is inherently multi-stage, we report both aggregate question-level accuracy and deployment-oriented image-level metrics.

Overall question-level performance Among all settings, the sequential SFT→GRPO achieves the best overall accuracy of 92.43%, outperforming Pure GRPO (91.49%), Pure SFT (89.41%), and zero-shot inference (17.15%). Although the gain over Pure GRPO is modest, it remains meaningful in a high-accuracy regime, where remaining errors are concentrated in ambiguous or cross-stage-sensitive cases.

Stage-wise trade-off between perception and reasoning

A closer inspection of the stage-wise results shows that SFT substantially improves the zero-shot model across all diagnostic stages, raising the overall accuracy from 17.15% to

89.41%. Pure GRPO further improves all five stages relative to Pure SFT, especially defect classification, localization, description, and process-informed analysis. Relative to Pure SFT, Pure GRPO increases defect description accuracy from 90.23% to 92.49%, process-informed analysis from 88.86% to 90.89%, and overall question-level accuracy from 89.41% to 91.49%. The sequential SFT→GRPO setting achieves the best overall accuracy and the strongest Q1, Q2, Q4, and Q5 performance, indicating that supervised schema stabilization and reinforcement alignment remain complementary.

Why the nested SFT→GRPO design works best The strongest aggregate performance is achieved by applying SFT and GRPO sequentially. Relative to Pure GRPO, the sequential SFT→GRPO setting improves overall accuracy from 91.49% to 92.43%, defect classification from 93.03% to 93.54%, defect description from 92.49% to 93.62%, and process-informed analysis from 90.89% to 92.45%, while preserving comparable localization performance. These results support the view that SFT provides schema stabilization and a strong supervised initialization, whereas GRPO refines tuple-level consistency and downstream reasoning under the same fixed five-question MCQ protocol.

Benchmark against lightweight generic VLMs Comparison with lightweight off-the-shelf VLMs highlights the difficulty of the benchmark. Qwen2.5-VL-3B reaches only 17.15% overall question-level accuracy, and InternVL2.5-1B achieves 16.60%. The large gap to the post-trained variants shows that wafer-map diagnosis is not solved by prompting alone. Instead, reliable performance requires semiconductor-specific post-training that explicitly couples morphology interpretation with structured diagnostic outputs.

Deployment-oriented error profile Table 4 complements the stage-wise results by showing that the best aggregate model is not necessarily the one that optimizes every deployment-oriented indicator in isolation. The different post-training strategies induce distinct operating points. Pure SFT attains the lowest fatal misses (FM= 11,677), but it also produces the highest false alarms (FA= 12,696) and the highest 0/5 failure rate. Pure GRPO slightly increases FM to 12,248 while reducing FA to 11,995 and eliminating 0/5 failures. The sequential SFT→GRPO setting achieves the strongest overall question-level accuracy and greatly reduces FA to 3,293, although its FM remains higher than the two single-stage post-training variants. This balance is especially important in fab settings, where the operating point must account for both missed excursions and unnecessary alarms.

Question-level versus image-level strict correctness Although the sequential SFT→GRPO setting achieves the highest overall question-level accuracy, Pure GRPO attains the highest strict 5/5 rate at 0.8524, showing that question-level and image-level metrics capture complementary aspects of reliability.

Table 3: Performance comparison on *WaferMapVQA*. Overall denotes aggregate question-level accuracy under the five-question protocol.

Model	Scale	Overall	Anomaly	Defect Tasks			
			Detection	Classification	Localization	Description	Analysis
Qwen2.5-VL	3B	17.15	48.78	3.19	11.69	1.52	20.58
InternVL2.5	1B	16.60	42.44	10.53	4.12	9.50	16.44
Pure SFT	3B	89.41	85.17	89.73	93.07	90.23	88.86
Pure GRPO	3B	91.49	85.24	93.03	95.81	92.49	90.89
WaferMapVQA (Ours)	3B	92.43	86.97	93.54	95.59	93.62	92.45

Table 4: Deployment-oriented reliability under the five-question protocol. FM counts abnormal wafers missed by the anomaly gate, and FA counts normal wafers incorrectly flagged as abnormal. Lower is better for FM, FA, and 0/5, while higher is better for 5/5.

Setting	FM↓	FA↓	5/5↑	0/5↓
Pure SFT	11,677	12,696	0.8425	0.0278
Pure GRPO	12,248	11,995	0.8524	0.0000
WaferMapVQA (Ours)	18,120	3,293	0.8376	0.0055

Ablation on Low-Budget Post-Training Strategy

We compare four settings to isolate the contribution of each post-training stage: (i) zero-shot inference, (ii) SFT on the full 5% post-training pool, (iii) GRPO on a nested subset drawn from that pool, and (iv) the sequential SFT→GRPO setting. Since all settings share the same five-question MCQ task, prompt protocol, and evaluation pipeline, the differences in Table 3 and Table 4 can be attributed directly to the training strategy.

Effect of zero-shot inference Zero-shot inference shows that generic vision–language capability is insufficient for reliable wafer diagnosis. Without task-specific post-training, the model fails to acquire stable semantics for defect classes, localization, appearance, and process-informed interpretation, indicating that off-the-shelf deployment is inadequate for fab-facing use.

Effect of SFT SFT on the full 5% pool yields a major improvement over zero-shot inference and stabilizes the structured action schema. It raises overall accuracy to 89.41% and achieves balanced performance across all five diagnostic stages. However, compared with GRPO-based settings, Pure SFT does not achieve the best stage-wise performance and remains less effective in optimizing deployment-oriented operating points.

Effect of GRPO Relative to Pure SFT, Pure GRPO improves overall accuracy from 89.41% to 91.49% and improves all five stage-wise accuracies. It also achieves the highest strict 5/5 rate and eliminates 0/5 failures, suggesting that reinforcement alignment improves tuple-level consistency. However, its false-alarm count remains high, indicating that answer-level consistency alone does not fully resolve deployment-oriented trade-offs.

Effect of sequential SFT→GRPO The sequential setting achieves the best overall question-level accuracy among all compared settings. This result shows that SFT initialization remains beneficial even after GRPO becomes effective. Starting reinforcement alignment from an SFT-stabilized policy yields the best operating point by combining the complementary strengths of the two stages: SFT provides schema stability and supervised initialization, whereas GRPO improves later reasoning stages and deployment-oriented behavior.

Supplementary case analysis of SFT vs. GRPO To better understand the SFT–GRPO trade-off, we compare the Pure SFT and Pure GRPO checkpoints at the question level. Across disagreement cases, GRPO corrects 28,954 SFT errors, whereas SFT retains 24,875 cases that GRPO misses. Most corrected cases are concentrated in downstream semantic stages rather than in the anomaly gate: Defect Description contributes 8,320 corrected cases, Defect Analysis contributes 8,306, and Defect Classification contributes 8,026, while Anomaly Detection contributes only 87. This pattern indicates that GRPO mainly improves morphology-conditioned semantic selection rather than merely output-format compliance.

Industrial interpretation From an industrial perspective, Pure SFT reduces fatal misses most strongly but suffers from many false alarms and the highest 0/5 failure rate. Pure GRPO improves tuple-level consistency and reasoning-stage accuracy, but still leaves a high false-alarm burden. The sequential SFT→GRPO setting provides the most practical overall compromise by achieving the best aggregate accuracy and substantially reducing false alarms relative to both Pure SFT and Pure GRPO.

Ablation summary Taken together, the ablation shows that zero-shot inference is inadequate for structured wafer diagnosis. SFT provides supervised initialization, GRPO improves tuple-level consistency and downstream reasoning, and the sequential SFT→GRPO achieves the best balance between accuracy and deployment reliability. These results validate the proposed two-stage post-training design as a practical low-budget recipe for semiconductor diagnostic reasoning.

Conclusion

We presented *WaferMapVQA*, a reinforcement-aligned sequential diagnostic planning framework for wafer defect diagnosis. By reformulating wafer-map analysis as a five-stage, perception-first diagnostic task, the proposed framework extends wafer inspection from single-label prediction to a more auditable and deployment-oriented setting.

Our study shows that low-budget post-training can substantially improve a small open-source VLM for semiconductor diagnosis. SFT stabilizes the action schema and provides a strong supervised initialization, while GRPO improves tuple-level consistency and reasoning-heavy stages. The best aggregate question-level performance is achieved by the sequential nested SFT→GRPO setting.

Experimental results show that different post-training strategies induce distinct deployment trade-offs. Pure SFT reduces fatal misses most strongly but suffers from many false alarms, while Pure GRPO improves strict tuple-level consistency but still leaves a high false-alarm burden. The sequential SFT→GRPO setting provides the most practical compromise by achieving the highest aggregate accuracy and substantially reducing false alarms.

More broadly, our findings suggest that lightweight reward-shaped policy learning may provide a practical alternative to large-scale agentic planning systems for industrial diagnostic reasoning. These results point toward trustworthy and deployable vision–language planning assistance for semiconductor manufacturing.

References

- Chen, J.; et al. 2023. Wafer map defect pattern detection and recognition based on convolutional neural networks. *Expert Systems with Applications*, 213: 118859.
- Saqlain, M.; et al. 2022. A deep learning-based framework for wafer map defect recognition and root cause analysis. *Expert Systems with Applications*, 197: 116743.
- Wu, M. J.; Jang, J. S. R.; and Chen, J. L. 2014. Wafer map failure pattern recognition and similarity ranking for large-scale data sets. *IEEE Transactions on Semiconductor Manufacturing*, 27(4): 537–546.
- Gu, S.; Lei, Z.; Wen, G.; Xu, Q.; Li, Z. S.; Chen, X.; and Yang, C. 2025. WamGLM: A multimodal large-scale language model for wafer map defect information in-depth query through multi-turn dialogue based on prototypical supervised contrastive learning. *Applied Soft Computing*, 185: 113962.
- Shen, H.; Liu, P.; Li, J.; Fang, C.; Ma, Y.; Liao, J.; Shen, Q.; Zhang, Z.; Zhao, K.; Zhang, Q.; Xu, R.; and Zhao, T. 2025. VLM-R1: A stable and generalizable R1-style large vision-language model. *arXiv preprint arXiv:2504.07615*.
- Pan, J.; et al. 2025. MedVLM-R1: Incentivizing medical reasoning capability of vision-language models via reinforcement learning. In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- Lin, S.-C.; and Lai, S.-H. 2025. CLIP-FSQAE: CLIP-guided finite scalar quantized autoencoder for few-shot anomaly detection. In *2025 IEEE International Conference on Image Processing (ICIP)*, 313–318.
- Lin, S.-C.; and Lai, S.-H. 2024. SQUAD: Scalar quantized representation learning for unsupervised anomaly detection and localization. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 166–182.
- Qwen Team. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.